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Photovoltaic Power Forecasting: A Review on Models and Future Research Directions

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ABSTRACT

The stability of power grid systems can be significantly affected by the unpredictability and volatility of power generation; however, accurate forecasting of solar energy power can help reduce this impact. This benefits the system through lower operating costs, balanced operation, and optimal dispatch. Over the past decade, extensive research has been published on this topic, exploring physical models, artificial intelligence (AI) techniques, and numerical and probabilistic approaches. Additionally, previous review studies centred their review discussions on a specific event horizon, others focused exclusively on the geographical horizon, and assessed only particular classes of photovoltaic (PV) output power forecasts. They paid little or no attention to other classes. Therefore, a thorough analysis of solar PV output power forecasting methods is required. In this paper, special focus is given to deep learning (DL), machine learning (ML), and hybrid methods, as these AI areas are gaining popularity. This study aims to provide a comprehensive and critical review of the latest AI applications. It also features a statistical analysis of forecasting errors based on over a hundred solar generation forecast studies. Additionally, the paper offers a brief introduction to the metrics used in ML, DL, and hybrid methods and their interpretation. A discussion of factors influencing forecasting errors is included. Future models will be more accurate because of the clarification that has been provided.

1 | Introduction

The world's power demand is rising, raising the possibility of a global energy crisis and its detrimental repercussions on the environment and human habitat. Given that global stocks of raw material—fossil fuels—have significantly decreased, the efficient use of energy has garnered substantial attention due to the serious social, political, and economic issues it raises. The ongoing advancement of human society and technology indicates a growing need for electrical energy [1]. However, the majority of the world's energy production methods still rely on fossil fuels, which account for more than 80% of worldwide consumption with an annual rate of consumption of 3.1 million tones of carbon, oil, and petrol. These sources are more in demand because of their low cost, but they also have detrimental consequences for the environment since they increase emissions of greenhouse gases, such as carbon dioxide (CO_2), which cause global warming [2]. With fossil fuels still the most extensively utilized source of energy, Figure 1 shows the historical and forecast global energy use. As a substitute for lessen the effects of greenhouse emissions, unconventional renewable energies first appeared around the close of the 1990s. The energy sources used in these systems are found in naturally occurring processes or materials that can spontaneously recycle and change back into energy on a regular or intermittent basis [3]. The growing importance of renewable energy is demonstrated by the integration of renewables into

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FIGURE 1 | Various forms of renewable energy.

public energy networks, the application of distinct approaches to optimise the prediction process, and the use of technologies like neural networks (NNs) for predicting energy production.

One of the most serious of these is the shift in the energy paradigm, which substitutes RESs for traditional and polluting energy production sources. As seen in Figure 1, there exist several RESs. These sources are more sustainable, cleaner, and need fewer resources. Additionally, these state-of-the-art technologies are already thought to be reliable and affordable. Depending on the region, different RESs are available in different nations. Solar energy is one of the most well-known RESs among them and is the primary source of hope [4].

Solar energy is a plentiful energy source since the sun emits vast amounts of it daily. It is produced by capturing the heat and radiant light from the sun. Due to its high power and low cost, solar PV has emerged as the most widely used solar energy technology worldwide. According to the quantity of installations, SPV power is now the most widely used electrical source globally; because solar cell efficiency is increasing and PV panel prices are continuing to decline. Additionally, PV may be able to help some developing nations deal with their power shortage problems by increasing solar PV system generation and reducing load-shedding and power outages [5]. The various forms of renewable energy sources are depicted in Figure 1.

Figure 2 shows the growth rates of solar power technology from 2000 to 2023. Between 2006 and 2009, the amount of PV energy produced worldwide is said to have tripled, and it has continued to expand significantly, reaching 100 GW in 2012. The total PV output in 2024 reached 1866.31 GW as presented in Figure 2.

Creating energy prediction models is a crucial activity that enables system optimization and extraction of the maximum amount of energy production. PV solar power systems are a primary means of producing renewable energy [6]. When light photons interact with semiconductor materials, the PV effect converts their energy into electrical power. Solar irradiance, incident radiation, and regional meteorological conditions like speed of wind, relative humidity, temperature, and are required for the construction of solar power generation systems [7]. The electricity generated by a PV system is a dynamic variable that changes over time because these environmental factors are naturally unpredictable.

Accurate PV power forecasting is essential to improving modern power networks' economic dispatch. Reliable forecasts greatly minimize supply uncertainty because solar power is inherently unpredictable and weather-dependent, allowing system operators to more effectively schedule generation units. PV forecasting directly lowers overall operating costs by reducing ramping requirements for conventional generators, minimizing reliance on expensive spinning reserves, and improving unit-commitment decisions. Additionally, accurate short-term projections provide better integration of distributed PV resources, congestion control, and effective real-time balancing. Consequently, a more thorough examination of the PV forecasting stage reveals that it serves as a strategic enabler for safe and affordable grid operation in addition to being a predictive tool.

The inability to estimate PV power output hurts market operations, planning, and scheduling activities, stability, and reliability in the electric grid [8]. As a result, predicting power generation is a primary area of focus for PV system research. The literature has developed several methods for estimating PV power, which can be divided into four main groups based on the forecasting horizon: short-term forecasting, which is defined as 48 to 72 h; medium-term forecasting, which is defined as a few days to a week; and long-term forecasting, which is defined as a few months to a year or longer [9, 10]. All of the predictions made by time horizons serve a specific purpose. While microgrid management, unit commitment, economic dispatch, system operations, power control, and other processes are all completed within a short timeframe, very short-term forecasters are utilized in the power market, PV system control, and the planning and maintenance of PV plants. One and multi-step-ahead predictors are taken into consideration in a different classification; the latter is more common in many applications [5, 11, 12]. For multi-step forecasting, there are three main approaches: the multi-input input multi-output approach, the direct approach, and the recursive approach. The primary factors influencing the performance of the predictors used to predict the amount of power generated by PV plants are weather, location, temporal resolution, quality of the data, and time horizon [13]. These variables allow for the selection of multiple predictors for a particular demand: physical methods that mostly use numerical weather prediction (NWP) models or satellite imagery that can be used to create geographic models [14]. In this case, long-term forecasting frequently uses meteorological data from satellite images. With NWP models, it is possible to forecast atmospheric conditions up to 15 days in advance without requiring any historical data [15]. These techniques can produce accurate results, but their main dependence is on weather stability. The setup of physical models is often rather hard due to the huge number of factors and costly resources that are required and not often available in some places of the world [16]. Furthermore, the initial hours of predictions for the majority of the available NWP are not very

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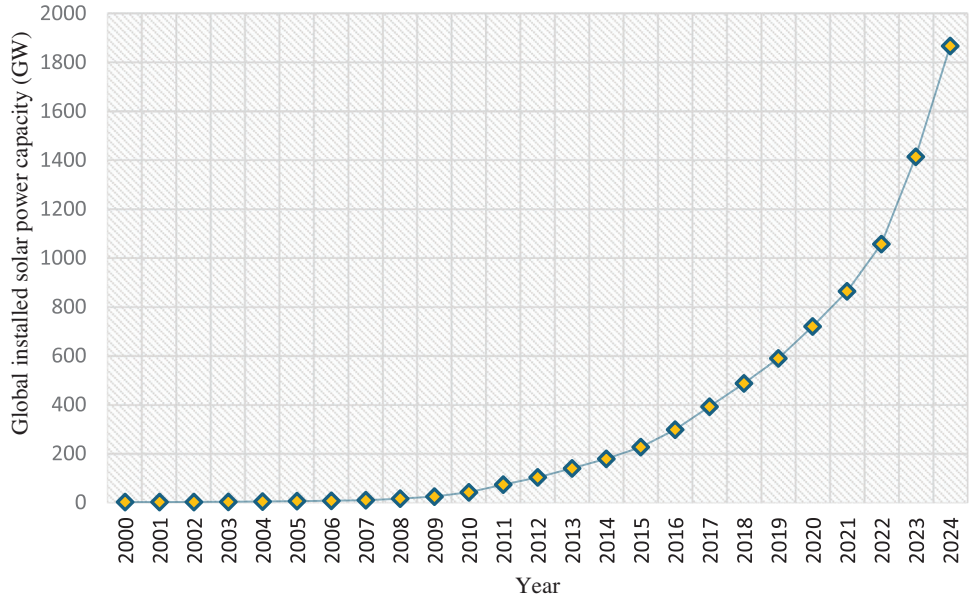


FIGURE 2 | Capacity of solar power installed worldwide, 2000–2024.

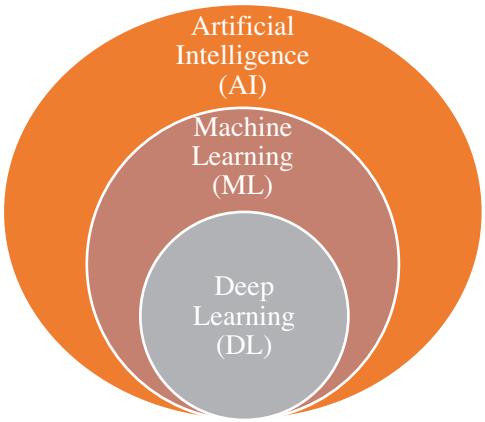


FIGURE 3 | Venn diagram presenting the relationship between AI, ML, and DL.

helpful for solar forecasts. Another option is to use statistical and probabilistic methodologies. Time series ensembles, exponential smoothing, regression models, autoregressive (AR) models, autoregressive integrated moving average (ARIMA), and probabilistic techniques are a few of these [17]. For projections that are made up to one day in advance, statistical methods are more suited. Several attempts were made to address the non-linearity issue. For example, [18] documents the widespread use of annual time series ensembles for PV forecasts. Venn diagram showing the relation between DL, ML, and AI in Figure 3.

Modern approaches such as support vector machines (SVM), extreme learning machine (ELM), k-nearest neighbors (KNN), artificial neural networks (ANNs), etc., are based on ML and AI techniques. These approaches, which do not require any PV system information, offer certain advantages that are common to statistical methods. They are primarily utilized for temporary applications and are employed when field measurements are accessible. A statistical or physical approach is combined with one of the sophisticated techniques mentioned above in hybrid approaches. Due to the integration of two effective strategies, this sort of technique offers very strong predictive accuracy. Three categories of models may be found in the literature: weather forecast data-driven models, on-site observation-based models, and weather forecast data-integrated models. These latter belong to the hybrid group and are gaining popularity. Recent studies on PV forecasting compare data-driven and DL approaches, which give greater accuracy but require large amounts of data and provide little transparency, with physics-based models, which provide interpretability but are unable to capture rapid cloud variability. While cross-regional models improve generalizability across climates, emerging hybrid and physics-informed models combine ML and physical principles to increase resilience. Uncertainty assessment is strengthened by the increasing focus on probabilistic approaches utilising Kullback-Leibler divergence (KLD), KS, and overlap coefficient (OVC) [19]. All things considered, hybrid and universal forecasting techniques seem to be the most successful in facilitating dependable and cost-effective grid dispatch.

The majority of current PV forecasting studies reveal significant research gaps concerning the types and amounts of data gathered. Numerous models rely on small single-site datasets that are frequently brief or devoid of high-resolution measurements of important variables like temperature, GHI, DNI, or sky photos. As a result, the model's generalizability is diminished, and its capacity to capture local variability is limited [20]. Additionally, several studies employ tiny or unbalanced training sets, which restrict the efficacy of DL and data-driven approaches. Few studies also use multi-source datasets at broad spatial and temporal dimensions, such as NWP, satellite imaging, sky cameras, and supervisory control and data acquisition (SCADA) signals. These gaps show that in order to create robust, transferable, and operationally dependable PV forecasting models, extensive, multi-site, multi-modal, and long-term datasets are required [21].

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Renewable energy forecasting has significantly benefited from advances in ML and DL. However, there are still several issues to address. The accuracy of forecasting models for renewable energy is affected by key factors such as input data selection, ML and DL algorithm choices, and missing data handling. To better understand the variables that influence renewable energy production, it is also important to develop reliable and interpretable models.

This study is distinct from others that have been published because its primary goal is to provide a broad review of the field of contemporary techniques. This paper provides the first thorough evaluation of the most applicable hybrid, DL, and ML approaches for PV power forecasts. When the model is quoted in the articles that are cited, these approaches are evaluated using external data, which is data that were not used in the model's construction. This guarantees that power can be predicted by the model with accuracy.

The structure of this paper is as follows: a quick overview of methods for AI, such as ML and DL, is given in Section 2. AI-based methods for forecasting PV power output are shown in Section 3. A literature review in each of the PV fields of interest is given in Section 4. Section 5 presents the key elements influencing the prediction of solar power. Concluding thoughts and future directions are provided in Section 6

2 | AI Methods

While there are numerous ways to define AI, 'imitating intelligent human behavior' is among the most widely used definitions. Systems included by this definition include those that possess the following characteristics: they can behave in a human-like manner, have a 'thinking process' akin to human mental processes, and are capable of acting in a human-like manner. Because of these factors, a variety of intelligent computer technologies are being utilized increasingly frequently in place of traditional methods [22]. The literature on PV power forecasts can be used to classify the many subfields of AI approaches into the following categories: ML is a research field of statistical models and methods that computer systems use to complete tasks effectively without the need for direct human intervention [23]. Computational intelligence (CI) studies the theory and development of computational paradigms influenced by language and biology. supervised, unsupervised, and reinforcement learning are subfields of ML; expert systems (ESs), let computers 'make decisions' by analyzing data and selecting from a range of options; and DL, also known as DL, which enables computational models with multiple processing layers to learn ways to represent information at multiple levels of abstraction [24]. Fuzzy logic (FL), genetic algorithms (GAs), ANNs, and are the three main pillars of computer-induced learning (CI). In many cross-disciplinary fields, including robotics, natural language processing (NLP), computer vision (CV), common sense knowledge and reasoning (CSKR), pattern recognition (PR), logic programming (LP), nonmonotonic reasoning (NMR), problem-solving and planning (PSP), and others, these main groups are frequently utilized for achieving particular tasks or are commonly utilized [25]. The literature has used several techniques to anticipate PV power [23]: Adaptive community optimization (ACO), particle swarm optimization (SWO), and other meta-heuristic approaches are included in GA. ML includes kNN, SVM, naïve Bayes (NB), linear regression (LR), decision trees (DT), and so on; NNs, such as radial basis function networks (RBFN), recurrent neural network (RNN), multilayer perceptron (MLP), etc.; and NNs, such as MLP, RNN, RBFN, and so on; FL and hybrid systems, which integrate two or more AI branches (such ANFIS, ANN-GA, for instance) [26]. According to Figure 1, the two most popular AI subsets for PV output power prediction are DL and ML. AI systems integrating complicated reasoning and representation learning will be the next big thing in PV forecasting. NNs (such as MLP, RNN, RBFN, etc.); hybrid systems and FL, which integrate two or more AI branches (such ANN, GA, ANFIS, etc). As seen in Figure 2, our analysis concentrates on DL, ML, and hybrid approaches, three of the most popular subsets of AI for PV output power forecasting. The next significant advancement in PV forecasting from an AI perspective will come from systems that integrate advanced reasoning with representation learning [27–28].

2.1 | ML

ML is defined by Arthur Samuel (1959) as techniques that allow computers to learn on their own from experiences (i.e., datasets) without explicit human programming. Figure 4 illustrates the four primary classifications of ML models: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. For PV power predictions, both unsupervised and supervised learning are frequently used.

Figure 5 presents a tree diagram that depicts the learning technique for the prediction of solar power. In the first case, an algorithm looks for correlations and dependencies between output and input properties. In unsupervised learning, the algorithm searches the dataset that is accessible for patterns and rules to better characterise the data; in this case, there is no output [29]. The reinforcement type is usually used for integrating data with high dimensions into lower dimensional data for analysis or display. Association and clustering are the two categories of problems that make up this learning domain. The technique of grouping items in a collection based on how similar they are to each other relative to the objects in the individual groups is known as clustering. Association rules, on the other hand, search large commercial databases for relationships between objects.

2.2 | DL

DL is the process of teaching computers to learn by giving them data, interactions, and observations with the outside world. The training of deep neural networks (DNNs) can be accomplished by DL, a relatively recent development in NNs programming, as standard NN-based techniques may be impacted by issues like overfitting, declining gradients, etc. NNs are a massive parallel-distributed processor composed of basic processing units with an innate tendency to store and make use of experiential information [30]. A typical neural NN consists of multiple layers of interconnected neurons, with each layer's neurons connected to the next. An input layer obtains a dataset and presents it to the NN; an output layer stores the network's response to the

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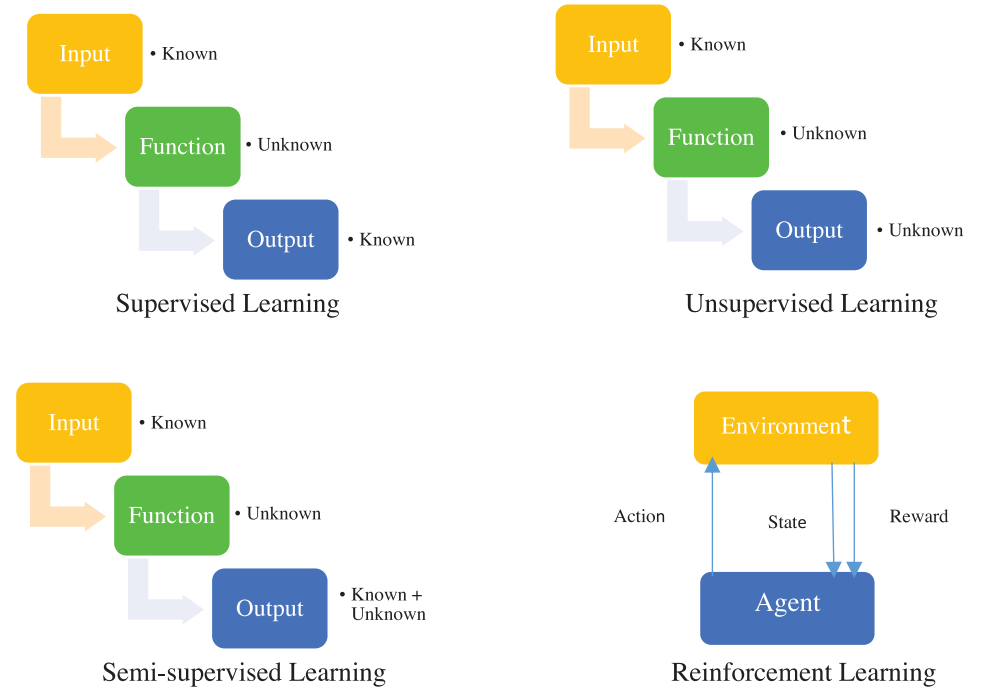


FIGURE 4 | Four main types of ML methods.

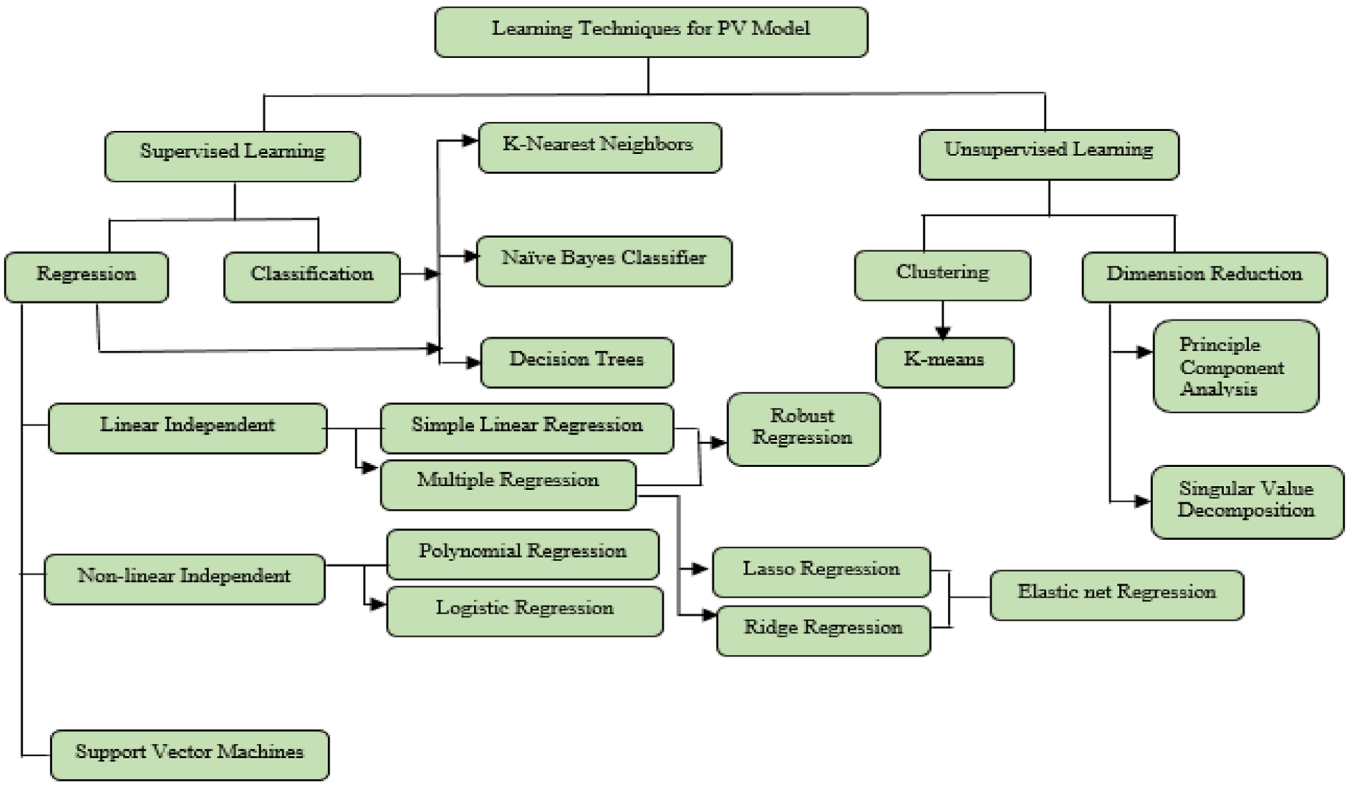


FIGURE 5 | An illustration of a learning technique tree.

input [31]. Between the input layer and the output layer, there could be one or more hidden layers. Basically, 'deep' refers to any NN having more than two layers. However, DL is a semi-supervised training method that works well with DNN training. The primary DL approaches are long long-short term memory (LSTMs), convolutional neural networks (CNNs), and other hybrid combinations, which are particularly useful for multistep forecasting [32]. The CNNs are particular NNSs that utilize a grid-like architecture instead of a general matrix multiplication [32]. Like ordinary NNs, CNNs include numerous hidden layers between the input and output layers [33]. The three layers that comprise the hidden layers are the pooling layer, rectified linear

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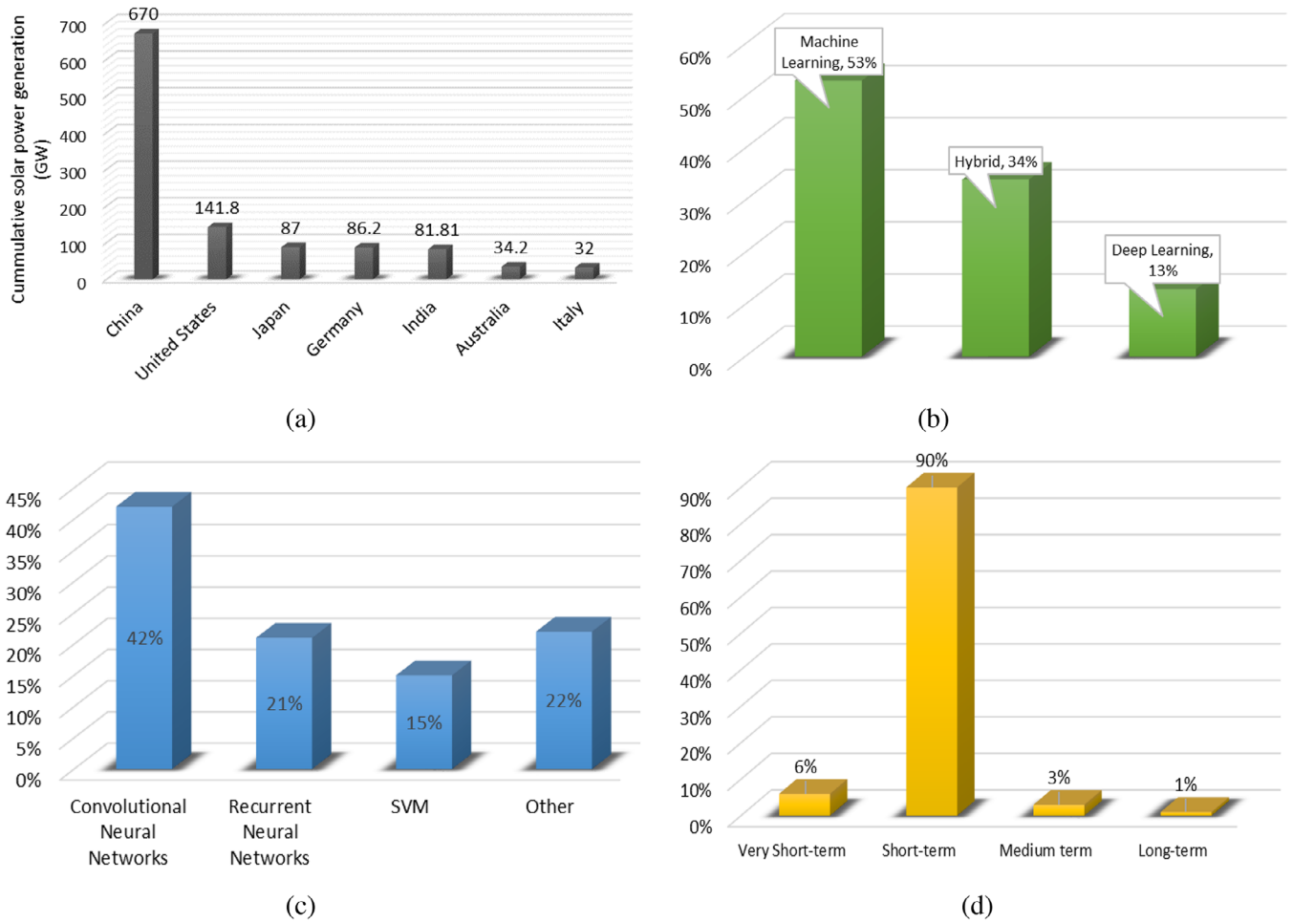


FIGURE 6 | (a) Nations that take part in PV forecasts; (b) AI-powered technique for predicting PV power; (c) most popular methods for DL; and (d) horizons of the time scale.

unit (ReLU), and convolution. Time series forecasting issues can benefit from the use of CNNs' inherent capacity to automatically identify patterns in raw input data and learn from them. A CNN model is often used to extract the most crucial details from a set of data by interpreting them as a one-dimensional image.

A memory cell, a forget gate, an output gate, an input gate, and the hidden state that all RNNs have are also features of LSTMs [34]. RNNs' primary flaw is that they are essentially incapable of managing long-term dependencies. RNNs cannot connect the data points as the distance between the input and output data points grows. To address this problem, several novel recurrent units (RU) have been developed in recent decades; the two most successful ones are gated recurrent units (GRU) and LSTM [35, 36].

3 | AI-Based Methods for Forecasting PV Power Output

The advanced DL and ML-based methods for forecasting PV output are covered in this section. The most widely used methodology in Figure 6, which comprises articles published in journals between 2015 and 2024, is ML approaches, which account for 53% of the total, while hybrid methods make up 34%. The most widely utilized DL strategies are ANN-based techniques, such as RBFN, RNN, and MLP (13%). Lastly, it may be said that 90% of studies have been done on the subject of long-term prediction, and Italian academics have been the most dedicated to this sector [37]. Over the past few years, the most popular forecasting method has been one-step ahead forecasting; however, due to recent developments in data analysis and soft computing techniques, multi-step forecasting has begun to gain traction [38–40].

3.1 | Utilising MLfor Forecasting PV Power

ML, which enables systems or programs to learn from datasets without requiring declarative programming, is a subset of AI. A thorough overview of the most pertinent studies on ML techniques for PV power forecasting for 2016–2024 is provided in Table 1.

Several ML techniques (such as the light gradient boosting machine (LGBM), eXtreme gradient boosting (XGBoost), LSTM, and linear model) have been evaluated by Buonanno, Amedeo et al. (2024) [41] for the combination approach. When compared to two baseline techniques based on numerical weather prediction, it was shown that the linear models were the most effective in forecasting solar production, with at least a 3.7%

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TABLE 1 | Lists the ML-based techniques used to anticipate PV power from 2016 to 2024.

Authors and ref.	Year	Technique	Resolution and time horizon	Parameters	Regional or point forecast	PV nominal power and region	Accuracy
Buonanno, Amedeo et al. [41]	2024	LGBM, eXtreme Gradient Boosting, LSTM, and linear model	5 min, 24 h ahead	Sunny day cloudy day, atmospheric temperature and cloud cover.	1 point, US	1.5 MW	*RMSE* = 3.7%
Ledmaoui, Younes et al. [42]	2023	XGBOOST, GAM, RF, DT, ANN, and SVR	5 min, 24 h ahead	Terrain elevation, Air temperature, optimum tilt of PV modules, global tilted irradiation at the optimum angle, diffuse horizontal irradiation, global horizontal irradiation, direct normal irradiation.	1 point	24 KWc, Benguerir city Morocco	RMSE = 2.6e−08 MAE = 0.00013 MASE = 0.29 R^2 = 0.99
Sedai, Ashish et al. [43]	2023	Ensemble models (RF, hybrid), DL models (GRU, LSTM, etc.), ML model (SVR), and statistical model (ARIMA)	1 day, 3 days, 5 days, and 15 days.	Temperature, wind speed, DHI, GHI, DN, power output.	Regional forecast	100-kW, Lubbock, Texas, USA.	RMSE 36%
Raj, Veena et al. (2023) [44]	2023	SVM, KNN, RF, and GBM	15 min, 24 h ahead	Wind velocity, relative humidity, ambient temperature, and solar insolation.	1 point	1.2 MW,	MAE = 19.8%, RMSE = 4.6%
Cantillo-Luna, Sergio et al. [45]	2023	LSTM-based tuned models, multi-layer perceptron, support vector regressor, random forest, and XGBoost,	15 min, 24 h ahead	Daily seasonality	1 point	1.5 MW, Valle del Cauca, Colombia.	MAE = 14.39% MAPE = 14%

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TABLE 1 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Parameters	Regional or point forecast	PV nominal power and region	Accuracy
Demir, Vahdettin et al. [46]	2023	KNN, ELM, GPR, and LSTM, SVMR	15 min, 24 h ahead	Min. relative humidity (RHmin, %), max relative humidity (RHmax, %), latitude (), month, longitude (), year, Monthly solar radiation (SR, MJ/m^2), elevation (m), avg. wind speed (WSavg, m/s), and max. temperature (Tmax, C), avg. temperature (Tavg C), min. temperature (Tmin, C),	Regional Forecast	Turkey	NSE = 0.875.
Pombo, Daniel Vázquez et al. [47]	2022	hybrid CNN–LSTM., LSTM, CNN, SVM, and RF	One to three days ahead	Temperature, cloud cover, relative humidity.	Point forecasting	10 kW PV array, Danish Technical University (DTU) located in Risø, Denmark.	RMSE = 7.5%
Zulkifly, Z. A et al. [48]	2021	Decision tree, Linear Regression, GPR, and SVM	January 2014 to 31 December 2015	Ambient temperature, solar irradiance, and historical PV power output.	1 point	5 Kw PV array, Universiti Teknikal Malaysia Melaka.	RMSE = 7.7(%) MAE = 35.2(%) R^2 = 92.3(%) MAD = 4.9
Rana, Mashud et al. [49]	2020	MLR, RF, SVR, and NNs,	1st January 2015 to 31st December 2016	Wind speed, precipitation temperature, cloud cover, and humidity.	Point forecasting	1.22 MW PV system in Australia.	MRE = 10.5% MAE = 110.46%
Su, Di, et al. [50]	2019	KNN, ELM, and SVR	1st March 2017 to 28 February 2018	Beam normal irradiation, diffuse horizontal, beam horizontal, and global horizontal.	–	406 MWp PV plant in the UK.	nRMSE = 6.74%
Amarasinghe, P. A. G. M., et al. [51]	2018	DBN, SVMR, and RF	2015–2017	Weather data and power generation.	Regional forecast	737 kW, Buruthakanda solar park, Sri Lanka.	RMSE = 0.03327 MAE = .00994 Bias = 00.00034

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TABLE 1 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Parameters	Regional or point forecast	PV nominal power and region	Accuracy
Persson, Caroline et al. [52]	2017	(GBRT)	5 min, 24 h ahead	Temperature, cloud cover, relative humidity (NWPRh) [%] precipitation (NWPPrecip) [mm] wind speed–west (NWPWsw) [m/s] wind speed.	Regional forecast	9:66 kWp, Nagoya Bay in Japan	RMSE = 0.100–0.137
Ahmed Mohammed et al. [53]	2016	Decision tree, gradient boosting, lasso regression, KNN	24-h ahead	Relative humidity, vertical integral of cloud ice water content, total column ice water, vertical integral of cloud liquid water content, and total column liquid water.	Point forecasting	European Centre	RMSE = 0.12973, MAE = 0.06211
Li, Zhaoxuan et al. [54]	2016	ANN and SVR,	15 min, 1 h and 24 h	Time, historical power information, meteorological forcasts.	Regional forecast	6 MWp, Florida	MBE = 0.03 MAE = 126.32 RMSE = 182.64 MBE = 0.0001 rRMSE = 0.410 MPE = 10.54
Jang, Han et al. [55]	2016	Support vector machine	15 to 300 min	Raw images, basic images, and processed images.	Point forecasting	South Korea	RMSE = 5.7367 MRE = 5.2825 R2 = 0.9745
Persson et al. [56]	2017	GBRT	1–6 h	Historical power generation data amd meteorological features.	Point forecasting at multiple sites.	Japan	RMSE = 0.100 and 0.137

Abbreviations: ANN, Artificial Neural Network; ARIMA, Auto-Regressive Integrated Moving Average; CNN, Convolutional Neural Network; DT, Decision Tree; GAM, Generalized Additive Model; GRU, Gated Recurrent Unit; LGBM, Light Gradient Boosting Machine; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; MASE, Mean Absolute Scaled Error; RF, Random Forest; RMSE, Root Mean Square Error; R^2, Coefficient of Determination; SVR, Support Vector Regression; XGBoost, eXtreme Gradient Boosting.

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root mean square error (RMSE) improvement. Despite problems with reliability and data availability in some regions of the world, precise resource forecasting has become essential to decision-making. Utilizing six ML algorithms— ANN, DT, RF, XGBOOST, generalized additive model (GAM), and support vector regression (SVR), Ledmaoui, Younes, et al. (2023) [42] developed a comprehensive and comparative analysis of solar power generation forecast in Morocco. The analysis was done using everyday of data from a PV plant that was built in Benguerir, Morocco, between January and December 2022. The models were evaluated after training and testing. This study employed four metrics to measure the model performances: R-squared (R^2), mean absolute scaled error (MASE), mean absolute error (MAE), and RMSE. With the lowest RMSE, MSAE, and greatest-squared values—all of which the ANN model recognizes as crucial performance criteria—the model's performance indicates that ANN is the most successful prediction model for power forecast [42]. Sedai, Ashish, et al. (2023) [43] offered a thorough performance examination of several ensemble, statistical, DL, and ML models for long-term solar power output prediction. As to the study [43], the RF model emerged as the most accurate of the four types of models examined, with 10% superior accuracy over multivariate models and 50% greater accuracy over univariate models. Raj, Veena, et al. (2023) [44] proposed ensemble-based ML models for estimating power generation from six distinct solar PV systems using random forests (RFs) and gradient-boosting machines. The models are based on a 1.2 MW grid-integrated solar PV power plant's performance over three years [44]. Cantillo-Luna, Sergio, et al. (2023) [45] presented an ML and DL-based approach for PV energy prediction that addresses these issues. It introduces the ConvLSTM1D strategy for this problem and incorporates RF, SVR, LSTM-based tuned models, multi-layer perceptron, and XGBoost. These models were evaluated using a variety of prediction horizons and a univariate time-series forecast of low-volume domestic solar power generation [45].

Demir, Vahdettin, et al. (2023) [46] used five alternative ML-based methods to estimate monthly sun radiation (SR). The models that are employed are Gaussian process regression (GPR), ELM, KNN, SVMR, and LSTM. These approaches had two stages to the modeling process [46]. Pombo, Daniel Vázquez, et al. (2022) [47] gave a case study of a PV system in Denmark and conducted a thorough evaluation of five distinct ML techniques: a hybrid CNN–LSTM, LSTM, CNN, SVM, and RF. The results consistently show that the most effective forecasters incorporate the indicated physics-informed traits, regardless of the particular ML approach and forecasting horizon [47].

Zulkifly, Z. A. et al. (2021) [48] created a ranking system for assessing the effectiveness of particular ML models. Four models—the Decision Tree, the LR, the GPR, and the SVM are examined in this work [48]. Rana, Mashud, et al. (2020) [49] developed a straightforward univariate technique to forecast PV output power several steps ahead. Maintaining high levels of accuracy for many step-forward predictions in ML algorithms is the goal. What makes the suggested strategy novel is when ML methods are combined with a data resampling methodology, using a single-step ahead forecasting on the re-sampled time series, ML approaches may calculate many steps ahead forecasts using a data resampling technique [49]. Although several ML-based PV prediction approaches have recently surfaced, the literature lacks a thorough evaluation of these techniques' effectiveness in a shared framework. Su, Di et al. (2019) [50] presented a thorough comparative analysis of the works on short-term PV prediction, comparing ten contemporary NNs and intelligent algorithms. A 406 MWp PV plant in the UK provides a one-year dataset for the proper fine-tuning and assessment of all approaches [50].

Amarasinghe, P. A. described [51] using various ML approaches for PV prediction at the Buruthakanda solar project in Hambantota, Sri Lanka. When the smart persistence (SP) approach and the implemented ML algorithms are compared for forecasting performance, the research reveals that the ML models perform better. Persson, Caroline, et al. [52] developed a non-parametric ML approach to predict solar power production over several sites from one to six hours ahead of time. A gradient-boosted regression tree (GBRT) model is constructed using historic power production data from 42 distinct PV rooftop installations and pertinent climatic factors. When compared to variants of the single-site linear autoregressive and GBRT models, the multi-site models perform competitively with respect to of RMSE over all prediction horizons. Three different ensemble probabilistic forecasting methods, based on seven different ML models, were provided by Ahmed Mohammed et al. 2016 [53] to generate solar power forecasts for the next twenty-four hours. Each ML model outperforms conventional standards such as autoregressive integrated moving averages, as we have seen, but the results obtained by the ensemble models are even more accurate than those obtained by any one ML model alone. In Florida, ANN and SVR are two frequently used techniques for forecasting the energy output of PV systems. Li, Zhaoxuan, and colleagues (2016) [54] assessed and contrasted these methods. A multilevel strategy is suggested using the ML algorithms that have been evaluated. The power measurements that were averaged over 15 min in 2014 provided the production data used in this investigation. Jang, Han, et al. (2016) [55] provided a model in this research that predicts solar power using an SVM learning approach and several satellite pictures. The vectors of motion for each using atmospheric motion vectors (AMVs) from satellite imagery and clouds are predicted. The SVM learning process is set up with a variety of input and output data sets by looking over four years' worth of historical satellite photos. Regarding prediction accuracy, we evaluate three models: the suggested SVM-based model, a conventional time-series model, and ANN. Similarly, a non-parametric ML method for multi-site solar power generation prediction over a forecast horizon of one to six hours is presented by Persson et al. (2017). A GBRT model is trained utilising historical power production and pertinent climatic variables associated with 42 distinct PV rooftop installations [56].

3.2 | Utilising DL for Forecasting PV Power

A subset of ML, specifically an ANN subset, is called DL. These ANNs can solve problems more effectively because their architecture has more processing layers. They can also extract more information from data. Prominent instances of DL use include face recognition and driverless driving. Table 2 presents the DL-based techniques used to predict PV power from 2017 to 2024.

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TABLE 2 | DL-based techniques used to predict PV power from 2017 to 2024.

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Sulaiman, Mohd Herwan et al. [57]	2024	Feed forward neural network (FFNN)	15 min	34-day	Irradiation, ambient temperature, and module temperature.	100-kW, India	RMSE = 374.40, MAE = 201, MAX = 2128.80
Mbey, Camille Franklin, et al. [58]	2024	MLP-LSTM-GA	15 min	30 years	Relative humidity, wind speed, solar tilt angle, temperature, and irradiance.	12 panels of 280 Wc each, Douala.	MSE = 1.25, RMSE = 1.11, MAE = 0.58, MAPE (%) = 0.04, R = 0.99
Gundu, Venkateswarlu et al. [59]	2024	JAYA-based LSMN (long short memory network)	–	2021	Solar Irradiation (SI), temperature(T), and solar power (SP).	100 kW, Asia	MAPE = 0.1210
Jamil, Irfan et al. [60]	2023	ANN, RNN, and CNN-LSTM	15 min	2015 to 2021	Relative humidity, wind speed, temperature, and irradiance.	100MW, Cholistan Desert in Pakistan.	RMSE = 154904.3
Elsaraiti, Meftah et al. [61]	2022	LSTM, MLP	5 min	(January 1, 2017, to December 31, 2017) + 2	Irradiance, wind speed, module temperature	Halifax, Nova Scotia, Canada	LSTM, MAE = 236.3, MAPE = 2.17, RMSE = 317.4, R2 = 0.93 MLP, MAE = 600.1, MAPE = 4.15, RMSE = 799.9, R2 = 0.99
Zhou, Hangxia et al. [62]	2021	Long short-term memory (LSTM), convolutional neural network (CNN),	7.5 min	October 2014 to September 2018	Phases, frequency, voltage, current, PV module temperature, and PV power.	Shaoxing city in the eastern part of China.	RMSE = 1.30, MAE = 0.70, MAPE = 18.82
Du Plessis et al. [63]	2021	GRU-RNN	5 min	2015-06-01 to 2019-05-31	Azimuth (Φ) sun position angles, altitude (β), month, hour, month data, and wind-direction.	75 MW	MAPE = 1.42%–8.13%
Lee, Chung-Hong et al. [64]	2021	Bi-GRU, Bi-LSTM, Bi-Simple RNN, GRU, and LSTM	5 min	1 November 2018 to 31 January 2020	Solar cell module's real-time data collector for power generation, temperature.	11.4 kV, southern Taiwan.	$R^2 = 0.890179$

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TABLE 2 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Qu, Yinpeng et al. [65]	2021	Temporal distributed model-gated recurrent unit pool (GRUP)	5 min	40-days	Temperature, relative humidity, data collector for real-time power generation of the solar cell module		NRMSE = 6.83%, NMAE = 4.12%
Mishra, Manohar et al. [66]	2020	WT_LSTM dropout method (wavelet transform long-shot-term-memory network)	10 min	2nd February 2016 to 30th October 2017	Altimeter, wind speed, atmospheric pressure, temperature, relative humidity, due point, visibilities, cloud coverage, and output of PVGS (solar photovoltaic power generating stations).	Urbana Champaign, Illinois.	RMSE, MAE, MAPE and R2 score are noted to be 0.0049, 0.0405, 0.02982 and 0.93817
Wang, Jianxing et al. [67]	2020	CNN-LSTM	30 min	2008	Wind direction, sun azimuth, wind seed, relative humidity, atmospheric pressure, ambient temperature, sun altitude angle, angle, azimuth 180 diffuse horizontal irradiance (DHI), global horizontal irradiance (GHI), etc.	Zhangbei	RMSE = 5.061 MAE = 2.871
Torres, José F. et al. [68]	2019	Deep neural network	30 min	1 January 2015 to 31 December 2016	PV, weather, and weather forecast	University of Queensland in Brisbane, Australia.	MAE = 109.52, RMSE 128.66
Zaouali, Kalthoum et al. [69]	2018	Auto-long short term memory (auto-LSTM), ARIMA, SVM	15-min	2015–2016	Solar power, weather data	4 PV panels delivering 1.3 kW.	LSTM, NRMSE = 0.138, ARIMA, NRMSE = 0.137, SVM, NRMSE = 12.37
Díaz–Vico, David et al. [70]	2017	Deep neural networks (DNNs)	10 min	Kaggle *AMS* 2013–2014	Solar radiation	Oklahoma's Mesonet	MAE = 7.43

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TABLE 2 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Jang et al. [71]	2024	CNN, biLSTM, MLP	24-h	2013–2022	Temperature (TMP), solar azimuth angle (AA), solar elevation (SE), relative humidity (REH), wind speed (WSD, wind direction (VEC), cloudiness index (SKY), and precipitation probability (POP)	100 kWp, Republic of Korea.	MAE (per site): Site 1.7 = 3.16, 3.33, 3.35, 3.34, 6.08, 1.72, 3.65, MSE (avg): 64.49, RMSE (avg): 7.60.

Abbreviations: ANN, Artificial Neural Network; Bi-LSTM, Bidirectional Long Short-Term Memory; CNN, Convolutional Neural Network; DL, Deep Learning; GA, Genetic Algorithm; GRU, Gated Recurrent Unit; FFNN, Feed-Forward Neural Network; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; NRMSE, Normalised Root Mean Square Error; R^2, Coefficient of Determination; RMSE, Root Mean Square Error; RNN, Recurrent Neural Network; WT, Wavelet Transform.

Due to the growing complexity, enormous volumes of data are needed, which means long computational times and challenging—even impossible—interpretability. Employing real solar power plant observations taken over 34 days at 15-min intervals, Sulaiman, Mohd Herman, et al. [57] used feed-forward neural networks (FFNNs) to estimate AC power output. An accurate forecast is achieved by recording the complex nonlinear relationship between module temperature, ambient temperature, and sun irradiation. A thorough and comprehensive study of the DL methods currently employed for smart grid applications, like solar energy production and electrical consumption projection, was presented by Mbey, Camille Franklin, et al. [58]. This analysis incorporates socioeconomic, demographic, and smart meter data into its long-term electrical consumption projections. PV power generation forecasts for the near future take climate variables, such as sun irradiance, humidity, and temperature, into account. Furthermore, we have put forth a brand-new hybrid DL technique founded on the ideas of GA, LSTM, and MLP.

Gundu, Venkateswarlu, et al. [59] employed multiple deep NN models in conjunction with various meteorological parameters to forecast solar electricity production. We assess the effectiveness of the different models and check the prediction model utilizing a 100 kW solar dataset collected in 2021 from a specific region in Asia. Here, the suggested approach uses a deep recurrent network model called the long short memory network (LSMN) for forecasting solar power. Jamil, Irfan, et al. (2023) [60] addressed this issue by concentrating on three DL approaches, including RNN, CNN-LSTM, and ANN. These methods are applied to solar energy variables (SEVs) to classify the best-predicted model, including performance ratio (PR%), soiling loss (%), and power generation (MWh).

In 2022, Elsaraiti, Meftah, et al. published a system for forecasting solar power plant generated electricity through the use of DL techniques. This is accomplished by evaluating the ability of a DL method relying on the LSTM technique to forecast solar power output. Performance evaluation and comparison of the MLP network with the LSTM network are done [61]. Zhou, Hangxia, et al. [62] presented a hybrid DL approach (training, forecasting, and clustering) that addresses the current issues with the solar power production prediction challenge by combining the wireless sensor network with attention mechanisms (AMs), LSTM, CNNs, and clustering techniques. To improve forecasting accuracy, Du Plessis et al. [63] presented an integrated inverter-level prediction model in 2021. LSTM, GRU, and feed-forward NN models based on DL are the contemporary NN models that apply these inverter-level, macro-level, and prediction methodologies. Multi-step forecasts based on data gathered for a real-world situation are provided 1–6 h in advance for a solar system rated for 75 MW. In 2021, Lee, Chung-Hong, et al. [64] substituted daily characteristics (24 h) for the typical seasonal features (365 days) to predict power generation based on historical monitoring data collected by ground-connected solar power plants. We also implemented a daily numerical projection of the generation of whole-point electricity. Preliminary experimental evaluations demonstrate the rationality of our developed approach and allow for the examination of the solar power forecast performance.

Qu, Yinpeng, et al. (2021) [65] introduced a unique hybrid model for distributed PV power production prediction, which

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is founded on GRUss. The proposed method uses historical PV power generation data from the local area to anticipate solar energy time series with intervals of 5 min one day in advance. Initially, it is advised to determine the daily variations in the solar power chain using a linear components extraction technique. Next, depending on the observed daily fluctuations in PV power generation, temporal distributed GRU models are built. To create a unique short-term solar power forecast model, Mishra, Manohar, et al. [66] used the wavelet transform (WT) concept and the capacity to train the LSTM-based deep DL technique. Before completing statistical feature extraction, the proposed WT-LSTM model applies the WT to partition the gathered solar power time-series dataset into discrete frequency series. Wang, Jianxing, et al. [67] introduced a hybrid forecasting approach utilizing mechanism modeling and DL in 2020. By eliminating subjectivity from the selection of model input variables, mechanism modeling precisely defines the primary weather variables employed in prediction. LSTM networks and CNNs are used to reconstruct the input features and comprehensively study the spatial-temporal coupling characteristics between meteorological variables.

Torres, José F., et al. [68] (2019) investigated the problem of predicting the electricity generated by PV solar systems at half-hourly intervals for the following day. The authors also introduced DL, a feed-forward NN-based DL technique that divides the forecasting challenge into many manageable problems for large-scale time series data. Also, extensive analysis using two years of Australian solar data, comparing the performance of DL with two distinct state-of-the-art approaches using NN and pattern sequence similarity, as well as assessing training time and accuracy. For multiple forecasting time dimensions, Zaouali, Kalthoum et al. (2018) [69] used an auto-configurable middleware using an LSTM technique to determine the critical timescale for the learning environment. Our DL model performs better than the SVM model during the recommended learning period, according to the results. However, in our opinion, the ARIMA performs better than our recommended auto-LSTM approach [69]. However, the ARIMA requires a longer execution time for forecasting 15 to 30 min in advance. In this study, Díaz–Vico, David, et al. [70], (2017) used DNNs to two issues: daily solar radiation forecast and wind energy. Numerical weather forecast systems, with their well-defined spatial organization, provide the inputs for these challenges. As we'll demonstrate, DNN ensembles and single deep models can produce better predictions than SVR. This ML technique is observed as the state of the art in regression. Similarly, for multi-site hourly day-ahead PV forecasting, Jang et al. (2024) [71] proposed a common model that combines a CNN → biLSTM feature encoder with a 2-layer MLP regressor [71].

3.3 | Utilising Hybrid Approches for Forecasting PV Power

The hybrid technique integrates two or more ML approaches to enhance performance and occasionally decrease the substantial data and processing needs. Concatenating two forecasting algorithms, fine-tuning the prediction model parameters, or altering the input data to facilitate easier interpretation are some ways to combine these methods. Table 3 presents the hybrid-based techniques used to predict PV power from 2015 to 2024. To improve solar power forecasting accuracy from solar farms and reliability for power system operators, Saxena, Nishant et al. (2024) [72] developed a hybrid ML model that integrates the SVM with KNN, an ML learning technique [72]. Furthermore, a comparison is presented between the performance of the conventional LSTM technique and the suggested hybrid approach. The proposed hybrid model is enhanced by dataset variation in SVM and structural diversity in KNN, respectively. The suggested method was tested for solar power forecasts using the Jodhpur real-time series information set, which was obtained from weather station data centers using Meteonorm. Khelifi, Reski, et al. (2023) [73] showed a study on the effective application of a novel hybrid technique for predicting short-term solar power output for four distinct Algerian PV plants. Within the suggested model, an essential regression is provided by an ELM in conjunction with a time-varying filter-empirical mode decomposition (TVF-EMD). The TVF-EMD approach is used to handle the variations in the dataset by dividing it into multiple more constant and stable subseries [73].

An LSTM and CNN are used in a hybrid model for stable power output forecasting that was given by Lim, Su-Chang, et al. (2022) [74]. The CNN detects meteorological conditions, but the LSTM develops power-generating patterns using climatic variables. The proposed model was trained and assessed using solar power production information obtained from a power plant located in Busan, Korea. Evaluations that were both qualitative and quantitative confirmed the model's efficacy. Dash, Deepak Ranjan et al. [75] (2021) examined a novel and effective hybrid forecasting method that combines robust minimum variance with empirical wavelet transform (EWT). Random vector functional link network (RRVFLN) includes a functionally extended direct link from input nodes to the output node and a random weight vector for the enhancement nodes. A unique hybrid technique called LSTM-GPR is proposed by Wang, Ying, et al. [76]. It combines GPR with long-term support (LSTM) to produce a highly accurate point forecast together with a dependable interval estimate. Two measures are used to compare the hybrid model to alternative methods: the reliability of interval prediction and the accuracy of point forecast. The advantage of the LSTM technique over conventional NNs is validated by numerical data. Using a hybrid ensemble average method that combines the best features of several ML algorithms into a single model, Nayak propose a unique hybrid ML approach, Aanchit et al. [77], 2020 for forecasting solar power generation. A univariate regression-based imputation method has been used to address missing values in insolation. VanDeventer, William et al. (2019) [78] submitted a support vector machine (GASVM) model for short-term power prediction of residential-scale solar systems. On GAs, the model is based. Before optimizing the evolutionary process through an ensemble technique, the GASVM model uses an SVM classification algorithm to classify past climate information. Alongside the PV system, a local weather station was created at Deakin University to accurately monitor the nearby surroundings and prevent errors due to the remote gathering of meteorological parameters (Bureau of Meteorology).

Bajpai, Arpit et al. [79], 2019 analysed several weather characteristics for PV forecasting and used many ML models. Various models' performances are compared. To compare the models with the 4 kWp PV device, error evaluation was used. The

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TABLE 3 | Hybrid learning based techniques used to predict PV power from 2017 to 2024.

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Saxena, Nishant et al. [72]	2024	KNN-SVM	10 min	1st January 2020, to 31st December 2020,	Hourly total power energy generation (HTEG), hourly average global radiation (HAGR), hourly average temperature (HAT), and hourly global radiation (HGR).	Meteonorm. India	Accuracy by 7.1 %, sensitivity by 6.62 %, and specificity by 7.25 %.
Khelifi, Reski et al. [73]	2023	TVF-EMD-ELM	5min, 30 min	2016	Solar irradiation, the temperature, and the angle.	35.1 MW, Algeria	RMSE = 18.38 nRMSE = 3.64 MABE = 13.63 nMAE = 2.70 R (%) = 99.80
Lim, Su-Chang et al. [74]	2022	CNN-LSTM	10 min	2020	NWP	16-kW, Busan, Korea	MAPE = 4.58
Dash, Deepak Ranjan et al. [75]	2021	EWT-RRVFLN	15 min	Jan 2020 to Dec 2020	Solar radiation, atmospheric temperature humidity.	1200 kWp, Alabama	MAPE = 1.501, RMS = E0.436, MAE = 0.292
Wang, Ying et al. [76]	2021	LSTM-GPR	5min	1 February 2016 to 28 October 2017	Wind speed, atmospheric pressure, humidity, cloud cover, temperature, atmospheric, and solar radiation intensity.	University of Illinois in Urbana-Champaign	RMSE = 264.98 MAE = 201.77 MAPE = 9.43

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TABLE 3 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Nayak, Aanchit et al. [77]	2020	An extreme learning machine (ELM), SVR with RBF kernel, a least square boosted decision tree model, a non-linear autoregressive neural network with exogenous signal (NARX-NN), and non-linear autoregressive neural network (NAR-NN)	5 min.	2016	Total power generated and insolation	1 MW, Gujarat, India	NAR-NN, MSE = 0.7089 RMSE = 0.8420 MAE = 0.5852 NARX-NN, MSE = 0.8105 RMSE = 0.9003 MAE = 0.6708 ELM, MSE = 0.4139 RMSE = 0.6434 MAE = 0.4272 LSBDT, MSE = 0.4508 RMSE = 0.6714 MAE = 0.4490 RBF SVR, MSE = 0.1666 RMSE = 0.4082 MAE = 0.2925 EA, MSE = 0.1295 RMSE = 0.3101 MAE = 0.1936
VanDeventer, William et al. [78]	2019	Genetic algorithm-based support vector machine (GASVM)	10 min.	278 days	Irradiance, temperature, and power	3 kW, Deakin University, Victoria, Australia	RMSE = 11.226, MAPE = 1.7052
Bajpai, Arpit et al. [79]	2019	SVR-RF	1-min interval	8 February 2013 to 10 October 2016.	Total cloud cover (%), snowfall amount (cm/m^2), relative humidity at 2 m above ground (%), solar radiation (W/m^2), precipitation (mm/minute), wind direction at 30 m height, wind speed at 30 m height (m/second), wind temperature at 2 m height (celcius), air pressure at sea level (hPa).	4 KW, Alfons-GoppelStr., Munich	RMSE = 109.48

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TABLE 3 | (Continued)

Authors and ref.	Year	Technique	Resolution and time horizon	Dataset	Parameters	PV nominal power and region	Accuracy
Eseye, Abinet Tesfaye et al. [80]	2018	Hybrid WT-PSO-SVM)	10 min	2014–2016	Wind speed, atmospheric pressure, humidity, cloud cover, atmospheric temperature, and solar radiation intensity.	5 KW, Beijing, China.	MAPE (%) = 5.5720, RMSE = 2.3146, SDE = 2.2076
Majumder, Irani, et al. [81]	2017	EMD-ELM	5 min	1 year	Irradiance, temperature, and power	SOA University, Bhubaneswar, Odisha	MAPE = 5.62, MAE = 0.0564, RMSE = 0.1058
Asrari, Arash et al. [82]	2016	BPSFLAANN	5-min	2006	Wind speed. humidity, and atmospheric temperature solar radiation	22MW, Florida	MAPE = 10.78%, WMPE = 8.8%
Ramsami, Pamela et al. [83]	2015	SR-MLR SR-FFNN SR-GRNN	10 min	3 February 2012 to 30 December 2013	Wind speed, wind direction, solar irradiance, rainfall, humidity, atmospheric pressure, and Daily air temperature	250 W, Newquay, Cornwall, United Kingdom.	SR-MLR, RMSE = 3.15 MBE = 0.33 MAE = 2.33 R2 = 0.914 SR-FFNN, RMSE = 2.74 MBE = 0.01 MAE = 2.09 R2 = 0.932, SR-GRNN, RMSE = 2.94 MBE = 0.14 MAE = 2.31 R2 = 0.926

Abbreviations: CNN, Convolutional Neural Network; ELM, Extreme Learning Machine; EMD, Empirical Mode Decomposition; GA, Genetic Algorithm; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; MAPE, Mean Absolute Percentage Error; PSO, Particle Swarm Optimisation; RF, Random Forest; RMSE, Root Mean Square Error; SVM, Support Vector Machine; WT, Wavelet Transform.

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accuracy is assessed concerning past weather information. The study suggests a unique method for PV forecasting in which pipelining clustering, classification, and regression techniques are used as the foundation for building a model. A hybrid predictive model known as Hybrid SVM-particle swarm optimization (PSO)-WT was introduced in 2018 by Eseye, Abinet Tesfaye et al. [80]. It integrates an SVM, PSO, and WT, for a one-day-ahead production power prediction of a microgrid solar power plant. To create the model, real power data from the PV system's SCADA system is combined with weather information from NWP for a year at an hourly time-step. The hybrid ELM and EMD forecasting approach has been mentioned by Majumder, Irani, et al. [81], 2017. Additional decomposition of the non-stationary time series yields separate (IMF). A short-term forecast is also included in this study to illustrate the model's accuracy. For hourly solar power prediction, a hybrid forecasting algorithm is proposed by Asrari, Arash et al. [82], 2016. The structure of the presented model incorporates both meta-heuristic optimization and gradient-descent optimization techniques, which balance the computational burden and prediction accuracy. Ramsami, Pamela, et al. [83] (2015) provided a novel method based on daily weather forecasts for predicting the 24-h stochastic energy output of solar systems. A performance assessment of the hybrid approach with both ANN models and LR models has also been published. Multiple LR NNs, feed-forward NNs, and generalized regression NNs were the first three single-stage models created. After selecting the more relevant input factors by stepwise regression, a hybrid modelling approach was employed. For small-scale PV systems or baseline comparisons, statistical and traditional ML techniques like ARIMA, SVM, and RF offer quick computation and comparatively low data requirements. However, their accuracy decreases under high weather variability due to limited nonlinear and spatiotemporal learning capabilities. Particularly in nowcasting and short-term horizons, DL techniques—such as LSTM, GRU, and CNN architectures—offer notable gains in learning intricate temporal dynamics and spatial cloud evolution. However, these models often have explainability issues and demand big historical datasets and high-performance computational resources. Superior performance for utility-scale and grid-level applications is demonstrated by more sophisticated hybrid models, such as CNN–LSTM, Transformer-based methods, and ML–NWP integrations, which show good generalization under harsh and quickly changing weather conditions. However, their reliance on multi-source data streams (such as sky photos and NWP forecasts), higher computing cost, and operational latency restrict their viability in contexts with limited resources. By measuring uncertainty and improving robustness across seasonal transitions, ensemble and probabilistic forecasting frameworks further enhance model reliability, making them appropriate for dispatch planning and market operations. Overall, even though deep and hybrid models have superior prediction accuracy, their practical implementation must carefully take into account the forecasting task's operational timeline, computing budget, model interpretability, and data availability.

4 | Review of the Literature in Various PV Fields of Interest

Numerous investigations employ multi-step physical modelling pipelines that apply atmospheric radiation models, transposition, and PV performance equations to translate NWP-based GHI forecasts into PV power. Because they offer clear, physically understandable forecasts and assist in distinguishing meteorological mistakes from PV system modeling flaws, these physics-based techniques are useful benchmarks. Furthermore, compared to solely data-driven models, they frequently generalize better under unusual circumstances, such as extreme weather. Crucially, these methods also make it possible to create hybrid models, in which AI methods add local corrections or residual patterns to physical forecasts. Hybrid frameworks enhance forecasting accuracy, robustness, and operational reliability across a variety of PV systems and environmental circumstances by fusing the flexibility of ML with strong physical applicability.

It should be noted that only current studies—those published between 2018 and 2024—have been included. A few remarks describe what the reader might expect from each paper in Table 4. Figure 7 shows the number of times a specific feature has been utilized in the examined publications to generate predictions.

5 | Key Elements Influencing the Prediction of Solar Power

The dataset quality is affected by several factors, including outliers, missing data, weather classification, and the range of forecasting horizons. Additionally, the model's performance declines during abrupt cloud occlusions, when rapid irradiance changes can significantly reduce prediction accuracy [100]. Therefore, by utilizing both short-term historical patterns and future cloud dynamics, hybrid data-driven and numerical weather prediction (DL–NWP) techniques demonstrate improved responsiveness in such situations. Seasonal transitions, including the shift from winter to summer irradiance conditions, often lead to notable changes in sun angles, atmospheric characteristics, and climate behavior. By incorporating uncertainty and reducing bias associated with a specific model structure, ensemble learning techniques and probabilistic forecasting frameworks enhance generalization during these seasonal fluctuations [101, 102].

A summary of the input variables influencing solar power forecasts is shown below.

5.1 | Forecast Horizon

The period between the actual and effective times of forecasting, or a future time frame for output prediction, is known as the horizon of predictions. As evidenced by Refs. [103, 104], some researchers have a preference for forecast horizon groups that include the short-, medium-, and long-term. Others have created a 'fourth' category, aptly called the 'very short-term forecast horizon', according to the guidelines for smart or microgrid decision-making. All classification criteria have not, however, been agreed upon by all parties. A few minutes or several hours are usually considered to be the short term. Using short-term PV predictions ensures the safety and reliability of the electricity grid. It's common to refer to medium-term forecasting horizons as one or a few days. PV power predictions for the medium term are used to schedule maintenance. The long-term forecasting horizon is generally over a week [105]. It can be applied to power system

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TABLE 4 | Review articles on power PV production predictions. Considered publication years: 2020–2024.

Year	Notes	Ref.
2024	This review paper examines ensembles, supervised learning, regression, physics-based approaches, and the popularity of AI methods in solar power forecasting, focusing on Neural Networks. This demonstrates a multimodal strategy to tackle the complex problems related to solar energy forecasts. Forecasting accuracy is further increased by including historical data, weather forecasts, and satellite imagery. Our work describes different error metrics for evaluating forecasting models.	[84]
2024	The analysis shows that the potential for producing electricity through photovoltaic (PV) technology may rise in some locations and fall in others in the future, though the potential for concentrated solar power (CSP) technology may drop globally due to various variables and show notable regional variations. Furthermore, there is significant uncertainty in projections, and the trends of wind resources fluctuate across different regions. As a result, numerous studies have adjusted wind speeds before estimating wind energy.	[85]
2023	This review focuses on solar PV energy production prediction models that can be used to verify daily planning and system operations in smart grids. Furthermore, the case studies and suggested examples are thoroughly examined, and the predictive techniques found in the studied literature are categorized based on the source of the input data. A comparison is made between the benefits, drawbacks, and contributions of the predictive probabilistic approaches. Lastly, recommendations are made for further research on short-term solar PV energy forecasts.	[86]
2023	An extensively updated assessment of hybrid and standalone ML algorithms for forecasting PV power is provided in this review paper. To meet the UN sustainable development goals by 2030 and ensure the sustainability of grid power, solar generation forecasting is essential.	[87]
2023	This review paper on prediction models of solar irradiance is based on deep learning. Gated recurrent units (GRU), CNN, RNN, LSTM, Attention Mechanism (AM), generative adversarial networks (GAN), and other hybrid models currently in use are some examples of these models. Our investigation shows that deep learning models outperform traditional models in solar prediction applications, particularly when combined with certain strategies that improve feature extraction. Moreover, deep networks particularly benefit from the application of data augmentation methods to enhance DL performance.	[88]
2023	An overview of recent work on ensemble methods for solar irradiance forecasting is given in this article. These methods can be broadly divided into two categories: competitive and cooperative ensemble forecasting. Additionally, while data variety and parameters are thought of as competing ensemble forecasts, pre-treatment and post-processing are viewed as collaborating ensemble methods for forecasting. This article examines each of these ensemble forecast techniques. Ultimately, a conclusion has been reached and suggestions for additional research have been covered.	[89]
2022	This study compares and examines many PV power forecasting strategies, including methodologies based on machine and deep learning. Additional techniques encompass group techniques, empirical techniques, time-series statistical techniques, and group techniques. Additionally, a variety of model parameter optimization methodologies are compiled, important variables influencing PV power predictions are examined, and input selections for models forecasting PV power production are presented. To address the challenges posed by contemporary grid systems, probabilistic forecasting is expected to be crucial in the PV power forecast that is required, hence this paper compares existing probabilistic and deterministic models for forecasting. The importance of data processing techniques that raise prediction accuracy is also underlined.	[90]
2022	To present a current understanding of the topic, this study reviews both established and new techniques. The field has benefited from knowledge transfer between wind and solar forecasting, which is reviewed. New potential are also noted, notably about the utilization of remote sensing technology. The research design and forecasting approaches are compared, and suggestions for excellent, repeatable outcomes are given.	[91]

(Continues)

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TABLE 4 | (Continued)

Year	Notes	Ref.
2021	This review paper presented a wide analysis of the many forecasting models existing in the literature. With a primary focus on short-term forecasts, the study provides a rigorous examination of the period of the dataset utilized in each model as well as a synoptic comparison of their performance metrics.	[92]
2021	This work examines studies that have been published in the last 5 years on deep learning-based prediction of wind and solar power. In addition to pre-processing methods, probabilistic and deterministic approaches, evaluation methods, and comparison methods, it goes into considerable detail regarding the datasets and data used in the examined works. To facilitate methodological comparisons, a tabular summary of the essential features of every assessed study is provided. Future research directions and the field's present issues are listed.	[93]
2020	This paper examines deep learning models that are utilised to process time-series data and forecast photovoltaic (PV) electricity and solar irradiance. For the discussion, we've chosen three stand-alone models and one hybrid model: RNN, GRU, LSTM, CNN–LSTM, and based on training period, weather type, season, prediction horizon, input data, and accuracy, the chosen models were compared.	[94]
2020	The research on PV power-generating output forecasting techniques is thoroughly explained in this paper. The study also investigates the mechanisms influencing the power grid's frequency stability when a significant amount of solar power is integrated. The new difficulty this poses is taken into account as it presents the present technical approach for PV power production to assist in system frequency regulation.	[95]
2024	This work describes the DTMGNet model, which uses a novel architecture that strikes a compromise between generality, robustness, and predictive fidelity to overcome data scarcity and heterogeneity in order to address the multi-site long-term PV forecasting challenge.	[96]
2025	This study assesses hybrid models for day-ahead PV power forecasting that combine physical and machine learning (ML) methodologies. Two hybridization strategies are examined: models directed by physics and ML informed by physics.	[97]
2022	For short-term photovoltaic (PV) power prediction, this paper provides a thorough benchmarking examination of physics-informed ML models. Support vector regression, random forest (RF), CNN, LSTM, and a hybrid CNN–LSTM are among the popular algorithms that the authors assess both with and without physics-based feature augmentation obtained from a typical PV performance model. The best overall performance is achieved by the RF with physics-informed inputs, demonstrating the usefulness of hybrid approaches for operationally dependable PV forecasting.	[98]
2021	This study evaluates several data-driven methods for day-ahead forecasting using NWP inputs, such as ensemble learners and neural networks, against a physics-based PV model. Overall accuracy is comparable, but the physics-based approach performs marginally better since it can generalize from a less amount of site data. The usefulness of hybrid forecasting techniques is supported by data-driven models, which capture local system behavior but require a large amount of training data.	[99]

management and maintenance. Long-term projections, however, have a low accuracy rate due to the complexity of the components that are taken into account [106].

5.2 | Weather Classification

Weather influence is one of the main sources of instability in solar power. The abrupt decrease in power generation that happens when the sun is veiled causes problems for grid dispatching [107]. Because of this, weather data are a major component of the forecasting model. Wind speed, humidity, atmospheric pressure, kind of cloud, quantity of sunshine, module temperature, and air temperature, are all factors associated with sunlight. Perveen et al. (2019) [108] examined how well the ANFIS-based model forecasts solar energy globally under various weather situations, including clear, hazy, hazy and cloudy, and cloudy skies, as well as for different climate zones, utilizing key meteorological data. If there are equal to or more than 9 h of sunshine, the sky will be clear (blue). Furthermore, the diffuse solar energy is either less than or equal to 25% of the total solar energy. If the sunshine hours lie in the range of 7 to 9 h. Furthermore, less than 50% or more than 25% of the world's solar energy comes from diffuse solar radiation, the sky will be hazy (fully). A partially hazy and cloudy

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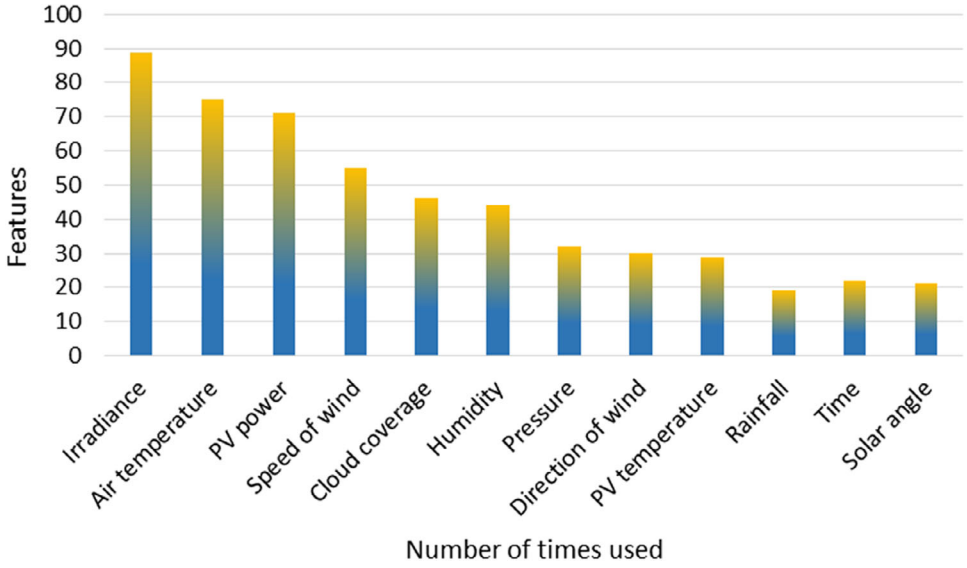


FIGURE 7 | The quantity of times a given feature has been used to make predictions in the papers that have been reviewed.

sky if the sunshine hour falls between 5 and 7 h. Furthermore, diffuse solar energy accounts for either more than 50% or less than 75% of the world's solar energy. A cloudy sky (fully) if diffuse solar energy is more than 75% of global solar energy and the sunlight hours are less than 5 h [108]. Similarly, Aman et al. (2025) [109] developed a revolutionary DL-based model for predicting solar power under various weather scenarios. The long-short term memory (LSTM) and CNN DL architectures are used in the proposed model's two-stage DL framework for precise solar PV power prediction. While the LSTM layer learns the patterns of solar power production that depend on weather variations to estimate PV power, the CNN layer's primary function is to identify weather conditions, such as sunny, partially overcast, and highly cloudy [109]. As a result, research has looked into the relationship between pertinent factors and solar power production. But the factor that has the biggest impact on solar power output is sun irradiation.

5.3 | Model Parameter Optimisation

PV forecasting models have made use of a range of optimization strategies; by choosing the right model inputs, forecast accuracy may be increased. By creating appropriate answers for various obstacles, optimizing algorithms can address a broad variety of issues. Good quality and efficiency can result from large-scale, ambiguous attributes, and high levels of problem complexity [110]. Numerous optimization strategies have been put out in different literature for PV power forecasts. These include the following: the firefly algorithm (FF) [111], the ant colony optimization (ACO) [112], honey badger algorithm (HBA) [113], a grid-search [114], PSO [115], the fruit fly optimization algorithm (FOA) [116], the charged system search (CSS) [117], the artificial bee colony (ABC) [118], and the GA [119], etc.

A GA works similarly to natural selection in that it selects the fittest individuals and eliminates the unfit ones to get the greatest results. Nonetheless, since a GA works with large amounts of data, it needs to do a lot of calculations. An ACO algorithm works similarly to ants that follow footsteps to find food. Ants use pheromones to communicate with one another and keep updating the optimal path while they look for food. ACOs would be more adept at locating the globally optimal solution because ants can communicate with one another through pheromones [120]. By studying how birds foraged for food, the PSO algorithm was created. A bird with memory and reference is represented by each particle. The local optimal solution problem is difficult to prevent because the PSO calculation process usually depends on experience [121]. Fruit fly foraging behaviour was examined in order to design the FOA. This approach is straightforward in the large dimensions, but iteration convergence is sluggish and data processing accuracy is low [122]. Parameters can be changed for optimization using the grid search technique. Extensive searches are conducted in each instance, experimenting with various factors in an attempt to find the optimal outcome. For a limited set of factors, this approach works well. Calculating with a lot of parameters would take a very long time.

By studying the ways the bees search for food, the ABC algorithm was devised. This algorithm's primary parameters are food supply, leader, scout, and follower. Food locations, storage capacities, collecting challenges, and other details are all retained in the leader's memory. The follower will have the chance to receive the information from it. To gather and update the food source data, the follower chooses which food sources to include. The scouter then updates the data on the honey source and looks for food establishments in the area. By iteratively going through the previous stages, each bee can identify the local optimal solution, which can then lead to the global optimal solution [123]. The fact that fireflies emit light and are attracted to one another led to the development of the FF. The firefly's position and brightness are first set by the algorithm. According to the definition, fireflies' mutual attraction is directly correlated with their luminosity, which is inversely correlated with distance. As a result, a firefly with greater brightness has the potential to draw in other fireflies. It moves randomly, though, if the brightness remains constant. As soon as the specified number of iterations is met, the firefly arrives at its destination. Coulomb and Newton's theories are utilized

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by the method known as the CSS, which is produced randomly. Every charged particle (CP) has its fitness value determined and is allocated a random place. After the highest fitness values are arranged systematically, the first 25% of the fitness values are chosen to be stored in the charge memory storage (CM). After that, the force that would be produced by each CP moving could be calculated. When the predetermined number of repeats is reached, this approach updates the positions and speeds of the CPs [124]. An ideal global answer can be found in the final fitness values.

5.4 | Project the Model Performance

The performance of the model estimation is necessary to assess how accurate it is in predicting. To find out how close actual values are to forecasted values, metrics are employed. The discrepancy between the actual and forecasted values is the error, which is the most fundamental statistic. To normalize this metric and make it unit-free, one can use the power plant's nominal capability, the real value (% error), or other techniques. Each of these measurements provides information regarding the type of error that was produced. MAE, MAPE, and RMSE are common instruments (see Table 5).

In solar forecasting, predicted values and accuracy are usually represented in terms of solar power output (kW) or irradiance (W/m^2). The most popular statistical metrics for determining the forecast's accuracy are briefly described here [125, 126]. RMSE, the average magnitude of the errors between the observed and forecasted values, is computed this way. It is especially useful since it provides a more complete measure of prediction accuracy by accounting for both smaller and larger errors. A lower RMSE indicates better predictability. The discrepancy between the predicted and actual numbers is averaged to get the MAE. It is a simple metric that provides insight into forecast accuracy. The MAE gives every error the same weight, regardless of its size, in contrast to the RMSE, which emphasizes greater errors as a result of the squaring process [127, 128]. Consequently, each error affects the MAE in the same way: Unlike other error metrics that consider the scale of the errors, the mean bias error (MBE) just looks at the direction of the errors—whether they are positive, or overestimations, or underestimates—and computes the difference between the actual and anticipated values. This shows that a prediction model will usually either underestimate or exceed the real values [129]. Relative RMSE (rRMSE) is a normalized version of the RMSE that is used to evaluate the anticipated accuracy of a model by accounting for the magnitude of the actual data. It is especially useful to assess a model's performance in relation to data variability when the data contains a variety of scales or units [130].

The normalised mean bias error (NMBE) and its corresponding measurements, NRMSE, NMSE, NMAE, and can be obtained by further normalizing all of these metrics. One specific example of NMAE is the mean absolute percentage error (MAPE), which is produced by normalizing the MAE using the actual value of the data series. Normalization allows for performance comparisons between PV power plants with different nominal powers and gives unit-free errors. The forecast of these statistical variables is frequently assessed using the forecast skill (FS) [131], also referred to as the performance skill (PS) or the forecast skill score (SS). It enables the user to contrast the forecasting approach that was produced with a technique of reference. The continuous ranked probability score (CRPS) is an additional statistic utilized in solar forecasting [132]. A large fraction of the predicted probability around its target value is rewarded by the CRPS to assess the forecast's accuracy and dependability. Just the most popular metrics from the list above are included here. Numerous other metrics are also employed, including the economic revenue (ER) [133], prediction interval normalised averaged width (PINAW) [134], the Brier skill score (BSS) [135], the NPMAE [136], skewness and kurtosis [137], and the correlation coefficient (R^2) [138]. Zhang et al. (2023) [139] developed a novel technique for probabilistic forecast reconciliation. By adding the KLD regularization term to the loss function, the reconciliation step becomes more adaptable and kind [139]. Similarly, a new distributionally robust chance-constrained (DRCC) evaluation technique based on Kullback-Leibler (KL) divergence was proposed by Shen et al. (2025) [140]. In addition, Cheng et al. (2024) [141] proposed a hypothesis testing method for evaluating the effectiveness of probabilistic seasonal rainfall forecasts based on the Kolmogorov-Smirnov (KS) one-sample test and the theory of probability integral transformation [141]. When estimating how similar two populations are in terms of their density functions, overlapping coefficients (OVL) are frequently utilized. Alhihi et al. (2023) [142] examined Pianka's OVC for two exponential populations in this work [142]. Pardo et al. (2025) [143] used LR with a potential Box-Cox transformation to create a covariate-specific OVL estimator [143]. In evolutionary and metaheuristic PV forecasting models, fitness functions measure the performance of each potential solution and direct optimization toward more precise forecasts. The difference between observed and predicted PV outputs is assessed using popular error-based fitness metrics like RMSE, MAE, MSE, and MAPE. Distribution-based fitness functions such as KLD, KS, and the OVC evaluate how similar the expected and actual distributions are for probabilistic forecasting. The optimization technique converges more quickly and guarantees dependable, resilient, and operationally significant PV forecasting performance by choosing suitable fitness criteria.

Each of the metrics is displayed along with its major properties and equation in Table 5. The variables, in this case, are the measured values (x_i), the predicted values ($\hat{x}_i$), the average true prediction value ($\bar{x}$), the number of samples (N), the cumulative distribution function (PDF) of the ith probabilistic forecast $F_i^f(x)$, the CDF of the ith observation $F_i^0(x)$, and the variance of the prediction distribution (σ). Nonetheless, some studies have stated that statistical error measurements alone are insufficient for evaluating the model's effectiveness.

Figure 8 shows the commonly utilized performance metrics for solar power forecasting.

6 | Conclusion and Future Trends

This paper delivers a detailed investigation of the most recent uses of AI approaches, like hybrid, ML, and DL, to predict solar power output. Although DL has been extensively researched for DL-based forecasting in general, DL-based PV power prediction has gotten relatively little attention thus far. The majority of

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TABLE 5 | Commonly used metrics for solar power forecast with their formulas and primary properties.

Metric (abbreviation)	Formula	Properties
Root mean square error (RMSE)	$RMSE = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \hat{x}_i)^2}{N}}$	Evaluate the forecasting models' overall accuracy. Square order, however, raises the rate of prediction inaccuracy.
Mean absolute percentage error (MAPE)	$MAPE = \frac{1}{N} \sum_I \left\lvert \frac{x_i - \hat{x}_i}{x_i} \right\rvert$	Assesses the forecasting models' uniform forecast errors as a percentage.
Mean absolute error (MAE)	$MAE = \sum_{i=1}^{N} \frac{\lvert x_i - \hat{x}_i \rvert^2}{N}$	Analyses model uniform forecast errors.
Mean bias error (MBE)	$MBE = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)$	Central tendency measure shown both over and under approximations.
Mean squared error (MSE)	$MSE = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^2$	Squaring and averaging
Coefficient of determination (R^2)	$R^2 = 1 - \frac{\sum_i (\hat{x}_i - x_i)^2}{\sum_i (\bar{x} - x_i)^2}$	Variability assessment shows how the observed data fits the model.
Normalized mean absolute percentage error (NRMSE)	$NRMSE = \frac{1}{\bar{x}} \sqrt{\frac{\sum_i (x_i - \hat{x}_i)^2}{N}}$	Evaluates the overall accuracy across a huge dataset. On the other hand, the forecast error rate rises with square order.
Normalized mean absolute percentage error (NMAPE)	$NMAPE = \frac{1}{N} \sum_{i=1}^{N} \frac{\lvert x_i - \hat{x}_i \rvert}{\frac{1}{N}\sum_{i=1}^{N} x_i} \times 100\%$	For long datasets, evaluate uniform forecast errors as a percentage.
Normalized mean squared error (NMSE)	$NMSE = \frac{1}{\bar{x}^2} \frac{\sum_i (\hat{x}_i - x_i)^2}{N}$	Calculation of the mean square of the errors. Greater significance is given to anomalies.
Skill score (SS)	$Skill\ Score = 1 - \frac{RMSE_{proposed}}{RMSE_{reference}}$	Represents an advancement over the baseline model, which is typically persistence.
Skewness	$Skewness = \frac{1}{N} \frac{\sum_i (\hat{x}_i - \bar{\hat{x}})^3}{\sigma^3}$	A measure of sharpness of the distribution.
Kurtosis	$Kurtosis = \frac{1}{N} \frac{\sum_i (\hat{x}_i - \bar{\hat{x}})^4}{\sigma^4}$	Measure of asymmetry of the distribution.
Continuous ranked probability score (CRPS)	$CRPS = \frac{1}{N} \sum_i \int_{-\infty}^{\infty} (F_i^f(x) - F_i^0(x))^2 dx$	Calculation of the difference between the actual value and the forecast range.
Correlation	$Correlation = \frac{\sum_i (x_i - \bar{x})(\hat{x}_i - \bar{\hat{x}})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (\hat{x}_i - \bar{\hat{x}})^2}}$	A measurement of the linear relation between two variables.
Kullback–Leibler divergence (KLD)	$D_{KL}(P\|\|Q) = \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} dx$	Measures how a reference distribution differs from a particular probability distribution.
Kolmogorov–Smirnov statistic (KS)	$D_{KS} = sup_x \lvert F_x(x) - G_m(x) \rvert$	The maximum vertical separation between two cumulative distribution functions is measured by KS.
Overlap coefficient (OVC)	$OVC(P,Q) = \int_{-\infty}^{\infty} \min\{p(x), q(x)\}\, dx$	Evaluates the degree of overlap or resemblance between two probability distributions.

research has focused on making predictions at specific locations, with minimal attention paid to regional models; thus far, no trustworthy general regional model has been published. Hybrid models fuse physical models with ML methods like ANNs and improve forecasting accuracy. Generally speaking, considering the following factors can help forecasters using AI algorithms become more accurate: Exogenous inputs, like cloud variation, should be considered: (1) Large, high-quality datasets are recommended; (2) prior treatment and database analysis are required to find missing outliers and data; and (3) in conjunction with other physical models.

DL models such as CNN and LSTM could be employed along with NWP models for long-term PV power PREDICTION. Some useful characteristics of these models and RNNs include their ability to assess the time-dependent nature of the data under study and perform a more thorough feature extraction. With the expansion of data (NWP models, satellite image data, onsite observations using SCADA), DL will become more and more helpful for PV power prediction. This is especially true for developing precise regional forecasts. Further research is required to understand DL-based forecasting techniques fully. Because there are more and more huge datasets and open-source programs (such as Keras, TensorFlow, Python, etc.) available, DL has become more and more popular, especially for time series forecasting. Conversely, ML-based models have demonstrated their potential to estimate PV power just like DL-based models, with the primary factors influencing accuracy being data quantity, quality, and learning methods. In contrast to ML-based models, DL-based models are more suited for huge databases. Very little research has been done

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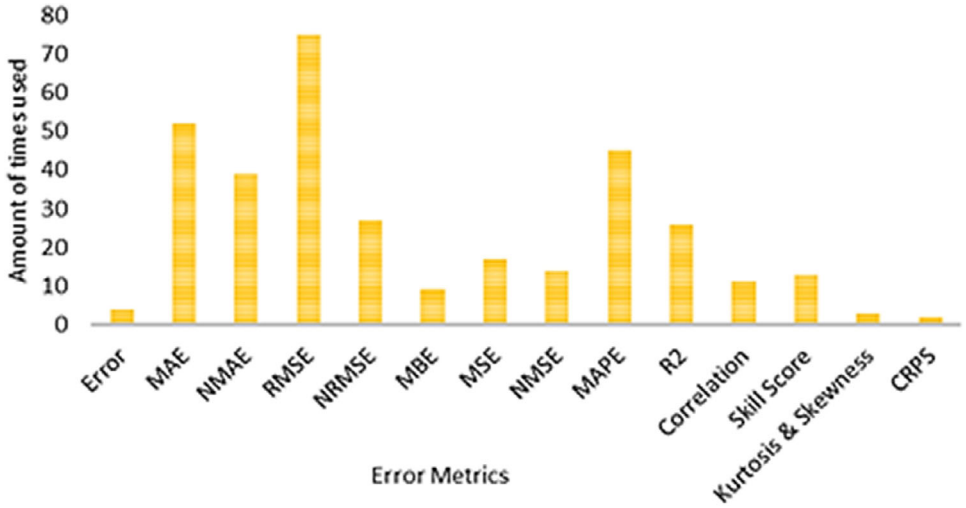


FIGURE 8 | Frequently employed performance metrics.

on improving the model's accuracy on cloudy days. Approaches for cloud cover quantification and categorisation predictions that leverage these features for DL models should result in notable improvements in accuracy. Seldom has the integration of DL-based techniques (like DCNNs) with physical models been studied. On the other hand, this kind of strategy might greatly increase forecast accuracy.

This study also employs over one hundred research papers for a thorough evaluation and statistical analysis of errors. Through quantitative analysis, we evaluated the distribution of the frequency with which certain parameters were applied in research investigations. (Includes forecasting horizon, system rated power, classifications of forecasting methodologies, quantity and kind of error measures, and forecast systems location). More studies on topics related to collective statistical evaluations, or meta-analyses, are, in our opinion, warranted. The RMSE and the MAE are the most employed metrics in the research field. In addition, KLD, KS, and the OVC are examples of distribution-based metrics that offer a more thorough assessment of solar power forecasts than conventional error measures. These metrics provide a deeper understanding of uncertainty, variability, and calibration by evaluating the degree to which predicted and actual power distributions match. OVC measures the degree of overlap between distributions, KS measures the maximum distribution deviation, and KLD quantifies information loss. When combined, they provide a more thorough evaluation of forecasting models, enhancing dependability and facilitating improved grid scheduling, planning, and integration of renewable energy.

Although popularity should not be the deciding factor for selecting a measure, it can be taken into account for comparisons. The system's attributes, including its location, the projected timeframes, and the study's goal, will determine which metric is best. It is often advised to use many measures of prediction error and to analyse the kind of error that was created. This will highlight the shortcomings of the suggested strategy to make it even better. For simpler comparison and comprehension, we also advise using normalised measures. There are currently not many studies on how to increase model accuracy on cloudy days. It is anticipated that forecasting techniques capable of estimating and classifying cloud cover would result in a significant improvement in accuracy when these parameters are used for DL models.

In the upcoming years, we aim to at least double or triple the quantity of reviewed papers. With this quantity, we may do a statistical analysis that is more precise in determining the various factors influencing the degree of prediction error. Recent research on production predictions in wind farms should, in our opinion, use data from our proposed criteria to assist the crucial analysis.

Nomenclature

AI	Artificial Intelligence
ANN	Artificial Neural Network
ANN	Artificial Neural Network
AR	Autoregressive
ARIMA	Auto-Regressive Integrated Moving Average
ARIMA	Autoregressive Moving Integrated Averages
BSS	Brier Skill Score
CNN	Convolutional Neural Networks
CRPS	Continuous Ranked Probability Score
DL	Deep Learning
DT	Decision Tree
DT	Decision Tree
ELM	Extreme Learning Machines
EMD	Empirical Mode Decomposition
GAM	Generalized Additive Model
GAM	Generalized Additive Model
GBRT	Gradient Boosted Regression Tree
GPR	Gaussian process regression
GRU	Gated Recurrent Units
IMF	Intrinsic Mode Functions

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KNN	K-nearest neighbours
KNN	K- Nearest Neighbour
LGBM	Light Gradient Boosting Machine
LGBM	Light Gradient Boosting Machine
LR	Linear Regression
LSTM	Long-Short Term Memory
MAE	Mean Absolute Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean absolute scaled error
MBE	Mean Bias Error
ML	Machine Learning
ML	Machine Learning
MLP	Multilayer Perceptron
MLR	Multiple Linear Regression
NNs	Neural Networks
nPMAE	Normalized Peak Mean Absolute Error
NWP	Numerical Weather Prediction
PV	Photovoltaic
R	Correlation Coefficient
R^2	R-squared
RBFN	Radial Basis Function Networks
RF	Random Forest
RMSE	Root Mean Square Error
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SVM	Support Vector Machine
SVMR	Support Vector Machine Regression
SVR	Support Vector Regression
XGBOOST	Extreme Gradient Boosting

Author Contributions

Upma Singh: conceptualization, data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Nisha Singh:** conceptualization, data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Hasmat Malik:** conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing – original draft, writing – review and editing. **Mohammad Asef Hossaini:** conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing – original draft, writing – review and editing. All authors contributed in the paper equally.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request

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